

Cloud Computing Services Reference

Service Models

Cloud computing delivers IT resources over the internet under three primary service models [NIST, 2011]:

Model	Full Name	Provider Manages	Customer Manages	Examples
IaaS	Infrastructure as a Service	Hardware, networking, VMs	OS, runtime, apps, data	AWS EC2, Azure VMs, GCP Compute
PaaS	Platform as a Service	IaaS + OS + runtime	Apps, data	Heroku, App Engine, Azure App Service
SaaS	Software as a Service	Everything	Configuration only	Gmail, Salesforce, Office 365
FaaS	Functions as a Service	IaaS + OS + runtime + scaling	Function code only	AWS Lambda, Azure Functions

Cloud Providers

The three largest public cloud providers hold over 65% of the global cloud market [Gartner, 2024]:

Provider	Compute	Container Service	ML Platform	Managed Database
AWS	EC2	EKS	SageMaker	RDS, DynamoDB
Microsoft Azure	Azure VMs	AKS	Azure ML	Azure SQL, Cosmos DB
Google Cloud (GCP)	Compute Engine	GKE	Vertex AI	Cloud SQL, Spanner

Cloud Database Services

Cloud providers offer managed database services that abstract away server provisioning and patching:

- **Relational (SQL)**
 - AWS RDS: managed PostgreSQL, MySQL, Oracle, and SQL Server instances
 - Azure SQL Database: fully managed SQL Server engine with built-in high availability
 - Google Cloud SQL: managed MySQL and PostgreSQL; automated backups and failover
 - All enforce ACID transactions and support read replicas for horizontal read scaling
- **NoSQL / Document / Key-Value**
 - AWS DynamoDB: serverless key-value and document store with single-digit millisecond latency
 - Azure Cosmos DB: multi-model globally distributed store with tunable consistency levels
 - Google Firestore: document-oriented NoSQL optimised for mobile and web applications
- **Cache Services**
 - AWS ElastiCache: managed Redis and Memcached for in-memory caching workloads
 - Azure Cache for Redis: fully managed Redis with optional geo-replication
- **Analytical / Data Warehouse**
 - AWS Redshift: columnar storage, petabyte-scale OLAP queries

- Google BigQuery: serverless analytics engine; separates compute and storage
- Azure Synapse Analytics: unified analytics with SQL pools and Apache Spark

The choice between SQL and NoSQL managed services mirrors the considerations for self-hosted databases: ACID compliance and relational integrity favour managed SQL; horizontal scalability and flexible document schemas favour managed NoSQL.

Machine Learning Platforms

Cloud ML platforms provide managed infrastructure for training and serving models at scale:

Feature	AWS SageMaker	Azure ML Studio	Google Vertex AI
AutoML	Autopilot	Automated ML	AutoML
Notebook environment	Studio Notebooks	Compute Instances	Workbench
Training job types	Spot, on-demand	Compute Clusters	Custom/pre-built jobs
Model registry	SageMaker Model Registry	Azure Model Registry	Vertex Model Registry
Serving	Real-time endpoints	ACI/AKS endpoints	Online/batch prediction
MLOps pipelines	SageMaker Pipelines	Azure ML Pipelines	Vertex Pipelines

Training large models in the cloud follows the same general pipeline as local training — data preparation, feature engineering, model selection, hyperparameter tuning, and evaluation — but distributes compute across GPU clusters and persists artefacts in object storage (S3, Azure Blob, GCS).

Cloud Networking

Cloud networking connects compute resources, storage, and external users:

- **Virtual Private Cloud (VPC)**
 - Isolated network segment within the provider's infrastructure
 - Public and private subnets segment workloads by security zone
 - Route tables and NAT gateways control ingress and egress traffic
- **Load Balancers**
 - Layer 4 (transport): routes TCP/UDP traffic by IP and port — AWS NLB, Azure Load Balancer
 - Layer 7 (application): routes HTTP/HTTPS traffic by path or header — AWS ALB, Azure Application Gateway
- **Content Delivery Networks (CDN)**
 - Cache static assets at edge locations close to end users
 - Reduce round-trip latency and offload origin servers
 - Examples: AWS CloudFront, Azure CDN, Google Cloud CDN
- **DNS Services**
 - AWS Route 53, Azure DNS, Google Cloud DNS
 - Support latency-based, geolocation, and failover routing policies

Deployment Models

Model	Description	Infrastructure Control
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Public	Resources shared across multiple tenants	Cloud provider
Private	Dedicated infrastructure for one organisation	Organisation or provider
Hybrid	Mix of on-premises and public cloud resources	Shared
Multi-cloud	Services distributed across two or more clouds	Organisation

References

- [NIST, 2011] P. Mell, T. Grance, "The NIST Definition of Cloud Computing," *NIST SP 800-145*, 2011.
- [Gartner, 2024] Gartner, *Magic Quadrant for Cloud Infrastructure and Platform Services*, 2024.